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Non-invasive profiling of the tumour microenvironment with spatial ecotypes

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Abstract | 摘要

Multicellular programs in the **tumour microenvironment (TME)** drive cancer pathogenesis and response to therapy but remain challenging to identify and profile clinically. BACKGROUND

Here, we present a **machine-learning framework** for multi-analyte profiling of spatially dependent cell states and multicellular ecosystems, termed **spatial ecotypes (SEs)**. By integrating **over 10 million single-cell and spot-level spatial transcriptomes** from diverse human carcinomas and melanomas, we identified **nine SEs** with broad conservation, each of which has unique biology, geospatial features and clinical outcome associations, including several linked to immunotherapy response. METHOD

Notably, **SEs were distinguishable by DNA methylation profiling** and were recoverable from **plasma cell-free DNA (cfDNA) using deep learning**. In cfDNA from nearly 100 patients with melanoma, **SE levels exhibited striking associations with immunotherapy response.** KEY FINDING

Our data reveal **fundamental units of TME organization** and demonstrate a **multimodal platform for profiling solid and liquid TMEs**, with implications for improved risk stratification and therapy personalization. IMPLICATION

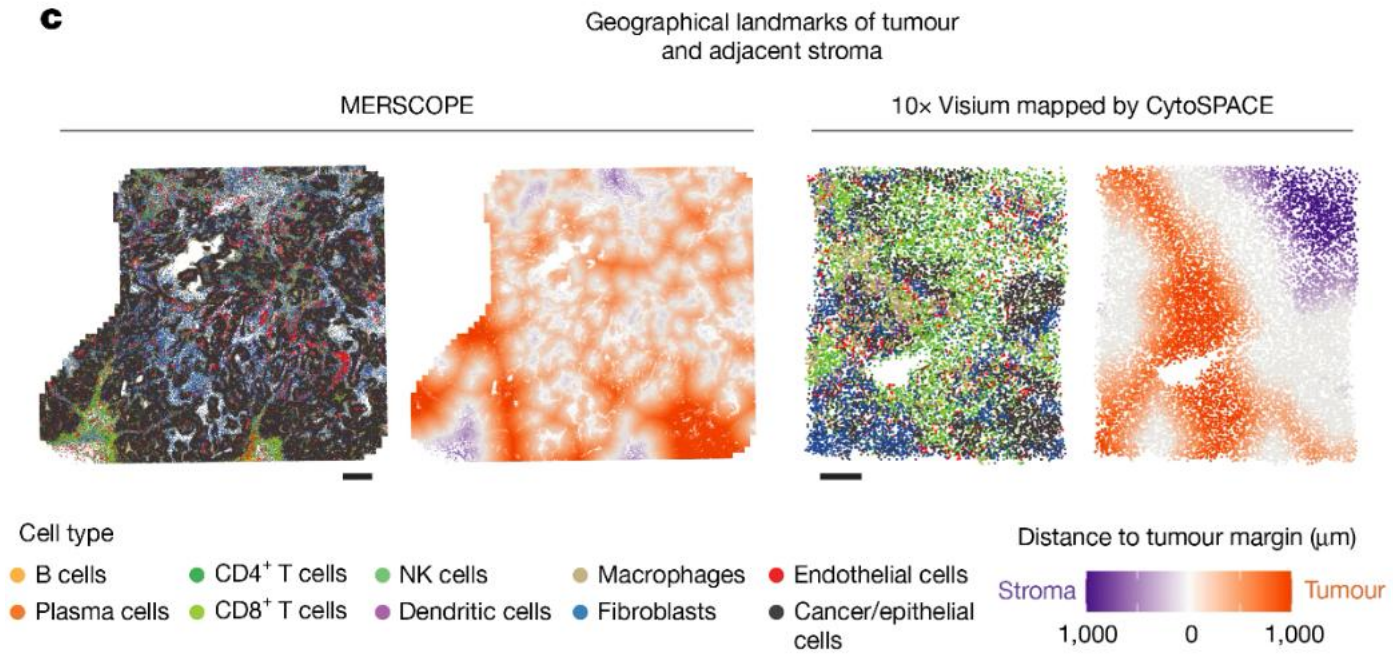
本文揭示了**肿瘤微环境空间生态型与临床结局的关联**，并实现了利用**血浆 cfDNA 甲基化信息非侵入性推断肿瘤空间生态型**，为通过液体活检评估肿瘤微环境及研究免疫治疗反应提供了新的研究思路。

Background | 背景

Beyond Cell Types: Spatial Organization of the Tumour Microenvironment

Tumour microenvironment (TME)

- 由恶性细胞、免疫细胞和基质细胞等共同构成。
- 不同细胞状态及其相互作用影响肿瘤进展、免疫逃逸和治疗反应。
- 传统的细胞类型分析不足以完整描述细胞之间的空间组织关系。



Background | 背景

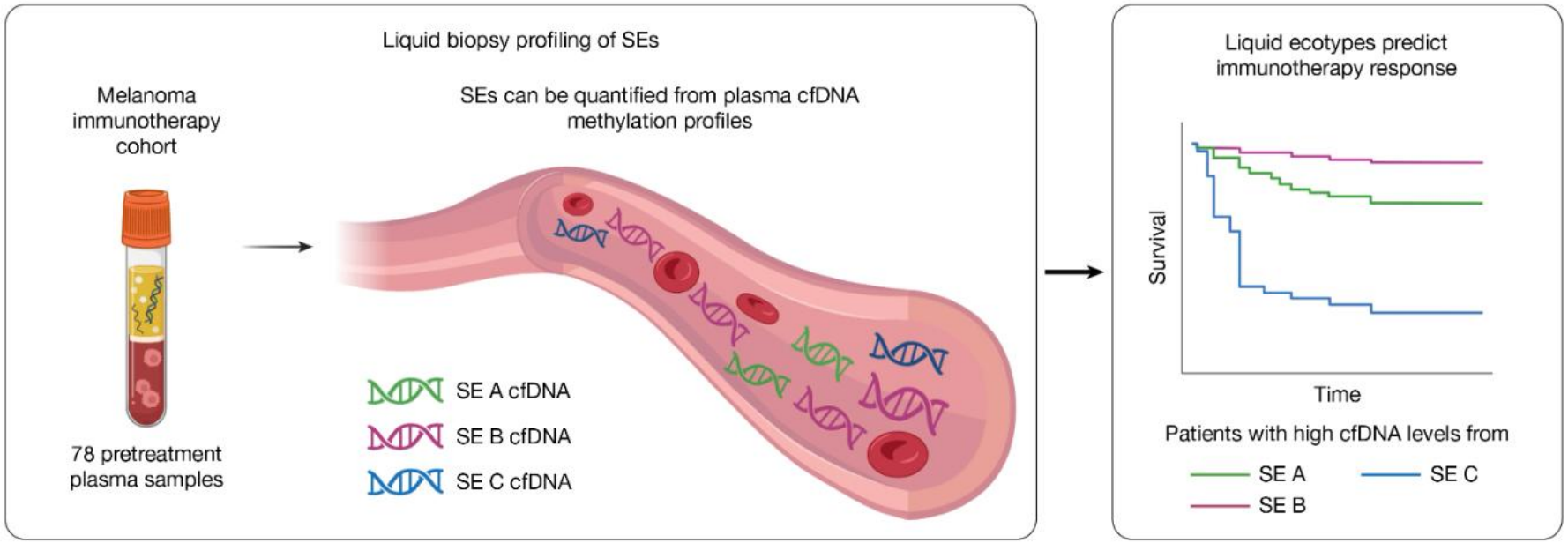
Current Methods Have Limitations in Spatial Ecotype Profiling

现有方法的局限	具体问题
预先定义标志物的成像方法	能检测的标志物数量有限，可能遗漏未知的细胞状态或生态模式
不考虑空间信息的生态型分析	可以识别细胞状态或生态型，却无法充分描述它们在组织中的空间组织关系
跨样本和跨平台整合能力不足	难以在不同样本、癌种和基因组平台间进行统一分析

Research gap
缺少能够系统识别空间依赖的多细胞生态模式，并整合不同癌种、样本及基因组平台的分析框架。

Background | 背景

From Invasive Tissue Biopsy to Non-invasive Liquid Biopsy



Study Rationale and Objectives |科学问题与研究目标

Ver.1

Central research question

能否建立一个整合空间组学与液体活检的分析框架，实现肿瘤微环境空间生态型的系统识别与非侵入性推断?

Three objectives

01

Identify spatial ecotypes

整合超过 1000 万份单细胞及空间转录组数据，识别具有跨癌种保守性和临床意义的 SEs。

02

Characterize clinical associations

研究 SEs 的生物学特征、空间特征及其与预后和免疫检查点抑制剂疗效的关系。

03

Recover SEs from plasma cfDNA

利用 DNA 甲基化数据与深度学习，评估能否从黑色素瘤患者血浆中推断 SE 水平，并与组织活检及临床疗效进行比较。

Study Rationale and Objectives |科学问题与研究目标

Ver.2

Central research question

能否建立一个整合空间组学与液体活检的分析框架，实现肿瘤微环境空间生态型的系统识别与非侵入性推断？

Three objectives

01 Identify spatial ecotypes

02 Characterize clinical associations

03 Recover SEs from plasma cfDNA

